

Single-cell foundation models represent genes in context, but reveal no context-specific gene function beyond co-expression

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Abstract

Background: Single-cell foundation models read a cell as a sequence of its genes and build, at every layer, a representation of each gene that depends on the other genes in the cell. Whether this contextual representation carries information about a gene's role in a given cell type, the single-cell counterpart of word sense in language models, has not been tested.

Results: We measured contextual gene representations in MaxToki (two sizes), scGPT and STATE, in 12 Tabula Sapiens cell types and an independent bone-marrow dataset. A gene's representation changes with cell type in a gene-specific way: the same gene's change agrees across independent cells at cosine 0.76, against 0.001 for a gene-shuffled control. The effect is strongest in the first two layers, survives rank control (99%), replicates on the independent

dataset (0.71), is partly learned (untrained model: 0.44 against 0.74 on the same cells) and does not grow consistently with model size. Genes that change more gain somewhat more from real context in next-gene prediction (partial correlation 0.16). The changes follow functional axes such as nuclear versus surface genes ($z = 17$ against random axes), but gene sets matched on size and co-expression give similar organisation ($p = 0.08\text{--}0.38$). Steering along the nuclear/surface axis, with large pushes, shifts the model's predictions for other genes ($p = 0.0081$); a second axis has no consistent effect and a third acts in reverse. Two tests for context-specific gene function, run in MaxToki-217M, were negative: transcription factors sit slightly closer to their curated targets than co-expression predicts, but not significantly more so where those targets are active. Functional organisation differs sharply between models ($z = 8.1\text{--}18.4$ for MaxToki, 3.2 for scGPT, 1.1 for STATE).

Conclusions: Single-cell foundation models build real contextual gene representations. In MaxToki, training organises them along functional axes no more than co-expression does, and we found no context-specific gene function beyond co-expression. Reading a gene's changed role in a new cell type will likely need models trained on more than expression. The functional organisation of contextual change offers a simple metric for comparing architectures.

Keywords: Single-cell foundation models; Contextual gene representations; Mechanistic interpretability; Gene co-expression; Gene embeddings; Transformers; Single-cell RNA sequencing; Model evaluation

Background

A recurring theme in the interpretability of large language models is that a network trained on a narrow predictive task builds internal representations of structure it was never taught. Sequence

models of Othello and chess build representations of the board and of latent variables such as player skill [1–3]; language models linearly encode space and time along identifiable directions [4]; and, in the best cases, these representations are not decorative but causally used, a claim one tests by intervening on the representation and measuring the effect on behaviour [5–7].

A closely related line concerns contextual representations. Contextual word embeddings assign the same word a different vector in every sentence, and the geometry of those vectors encodes word sense: “bank” near a river and “bank” in finance occupy separable regions [8–10]. These measurements need care, because representation spaces are strongly anisotropic (vectors cluster in a narrow cone rather than spreading evenly, so any two look similar) and a handful of “rogue” dimensions can dominate cosine similarity [9, 11–13].

Single-cell foundation models (SCFMs) offer a clean biological version of the same question [14–17]. They treat each gene as a token and each cell as a set or sequence of genes, either ordered by expression (Geneformer) or paired with binned or continuous expression values (scGPT, scBERT, scFoundation), and are trained to predict masked or next genes or values, in effect learning which genes tend to be active together. Their gene token embeddings (one fixed vector per gene) have been studied [15, 18]; their contextual gene representations (a gene’s vector inside a particular cell) are used as features downstream [16, 19] but have not been characterised as objects in their own right. Yet this is exactly where a model could, in principle, represent that a gene plays a different role in a macrophage than in a T cell: the biological analogue of word sense, and an open direction that the “cell as a token” framing invites but does not test [20].

Related work. Our methods descend directly from the language-model literature on contextualised embeddings. Ethayarajh [9] showed that contextual representations are anisotropic in all non-input layers, that same-word self-similarity falls in upper layers, and that every cosine

must be corrected against an anisotropy baseline. Timkey and van Schijndel [13] showed that a few rogue dimensions dominate similarity and that a scalar correction is not enough; Mu et al. [11] and Cai et al. [12] give the standard postprocessing and the local-isotropy picture. Coenen et al. [10] and Garí Soler and Apidianaki [21] show that word senses form separable clusters and that a word’s degree of polysemy is itself readable. We import both the measurements and the controls. From the literature on emergent, causally used latent variables [1–4] we adopt the evidentiary standard that a probe shows only correlation, so a causal claim needs an intervention against a matched control, together with its steering methods [5, 6], of which the extraction and amplification of interpretable feature directions in production models is the large-scale instance [22].

Geneformer [14], scGPT [15], scFoundation [16], scBERT [17], UCE [23] and STATE [24] differ in tokenisation (rank-ordered gene identity, binned or continuous expression values, or gene tokens derived from protein-sequence embeddings), and the review of Bian et al. [19] defines the “contextual gene embedding” that our work dissects. MaxToki [25], the primary model here, is a Llama-architecture, Geneformer-lineage model at 217M and 1B scales. Two earlier analyses come closest to a “context changes the gene” claim. scGPT reports cell-state-specific gene networks read from attention [15], but as a pairwise network, not as a property of the representation vector, and without a decomposition or a null. Geneformer’s in silico deletion measures how the embeddings of the other genes shift when one gene is removed from a cell [14], which is a change of context by one gene. Neither characterises how a gene’s representation differs between natural cell types. The geometry of gene representations in these models has been mapped, both for static gene embeddings [18] and for layer-wise hidden states averaged over cells, including their spectral structure [26] and the topological and geometric structure they do and do not encode, per tissue [27]. Residual-stream gene vectors built separately in several

tissues carry a small, tissue-dependent amount of transcription-factor-target signal beyond expression covariates [28]. How much, and how gene-specifically, a single gene’s vector changes from one cell type to another, the object here, has not been measured.

A critical literature finds that SCFM’s often fail to beat simple baselines: on zero-shot tasks [29], under deeper evaluation [30], on perturbation prediction [31, 32], and across utilities [33]. Sparse autoencoders find organised biological knowledge but minimal regulatory logic in these models [34], and edge scores read from SCFM attention add nothing beyond simple gene-level statistics for predicting which genes respond to a perturbation [35]. Co-expression is the canonical guilt-by-association signal [36], and some gene-embedding methods, such as Gene2vec, are built directly on it [37]; others build gene embeddings from text descriptions of genes instead [38]. Genuine context-dependent gene function exists. Some of it acts at the protein level, as in moonlighting proteins [39], and leaves no expression signature. Some acts through transcription, as when lineage-determining transcription factors select different enhancers with different partners by cell type [40].

Aim. We organise the question into two levels. Level 1 (a fact about the model): is a gene’s representation reshaped by context in a structured, gene-specific way, and is that reshaping learned and used? Level 2 (biology): does the reshaping reveal something about the gene’s role in each context beyond what co-expression already tells us? We answer Level 1 for MaxToki-217M with a rank control, an independent replication, a random-weights control, a link to the model’s predictions and a steering test, and compare contextualisation strength and functional organisation across four models. For Level 2, we test the functional organisation of the contextualisation against co-expression nulls and run two direct tests for context-specific gene function, all in MaxToki-217M. The study makes three contributions: (1) the Level 1 characterisation; (2) the Level 2 boundary; and (3) contextualisation as a model-characterisation metric.

Methods

Models

MaxToki-217M and MaxToki-1B [25, 41] are Llama-architecture autoregressive models over rank-ordered gene tokens, with 11 and 20 transformer layers and hidden widths of 1232 and 2304. We used the Hugging Face safetensors checkpoints through the transformers library with `output_hidden_states=True`, in float32. For MaxToki, “layer k ” is the k -th hidden state: layer 0 is the embedding output and layer $k \geq 1$ is the output of the k -th transformer block; the last hidden state (layer 11 of MaxToki-217M) is taken after the model’s final normalisation. scGPT [15, 42] (whole-human checkpoint; 12 layers; width 512) and STATE [24, 43] (the SE-600M state-embedding model; 16 layers; width 2048) were used for the cross-architecture comparison. STATE’s gene tokens are fixed protein embeddings from the Evolutionary Scale Modeling 2 (ESM2) language model [44], passed through a learned projection together with each gene’s expression. For scGPT and STATE, “layer k ” is the output of the transformer module with index k (the $(k+1)$ -th block). As an untrained control, we built a LlamaForCausalLM from MaxToki-217M’s configuration with random initial weights (the library’s default initialisation, seed 0) and the same vocabulary.

Datasets and annotations

The main data are the Tabula Sapiens [45] immune, kidney and lung datasets from CZ CELLx-GENE Discover [46–49]. For the immune dataset we used a 20,000-cell subset; its cell identifiers are listed in the code repository. These files store log-normalised expression ($\log_{10}$ of counts per 10,000) in their main matrix and the original integer counts in their raw matrix. All MaxToki analyses and the co-expression matrix read the raw counts; the scGPT and STATE in-

puts are described below. Contexts are cell types: the 12 largest cell types with at least 250 cells (macrophage, pulmonary alveolar type 2 cell, kidney epithelial cell, capillary endothelial cell, CD4-positive alpha-beta T cell, CD8-positive alpha-beta T cell, B cell, basal cell, pulmonary alveolar type 1 cell, intermediate monocyte, neutrophil and classical monocyte), with 1000 randomly chosen cells per cell type.

For independent replication we used all 5780 cells of the human CD34+ bone-marrow data of Setty et al. [50] as distributed with scVelo [51, 52], which holds velocity spliced and unspliced integer counts for 14,319 pre-filtered genes (11,842 in the MaxToki vocabulary); its main matrix holds the spliced counts only. Because the Tabula Sapiens counts include intronic reads, the main Setty analysis ranks genes by the sum of spliced and unspliced counts; a robustness run used spliced counts only. Contexts are the nine differentiation clusters with at least 130 cells: two haematopoietic stem cell (HSC) clusters, precursors, common lymphoid progenitors (CLP), two erythroid clusters, two monocyte clusters and dendritic cells (DC). Megakaryocytes (61 cells) were excluded. All cells of each kept cluster were used.

Gene Ontology (GO) annotations [53, 54] were taken from the gene-to-GO file distributed with GEARS [55, 56]. A gene belongs to a functional pole only if it is annotated directly to one of the listed GO terms; annotations were not propagated through the GO hierarchy. Curated transcription-factor (TF) targets were all human TRRUST v2 interactions [57, 58] together with DoRothEA interactions of confidence A or B [59, 60]. Gene symbols were mapped to Ensembl identifiers with the Geneformer dictionaries (version gc104M) [14, 61].

Input encoding and extraction of contextual gene representations

MaxToki uses Geneformer’s rank-value encoding [14]: a cell’s counts are normalised to the cell total, each gene is divided by its median value from the model’s gene-median dictionary, and the

159 non-zero genes are ranked in decreasing order. A per-cell scale factor does not change the order
160 within a cell, so the order is that of raw count divided by gene median. The first 1022 genes
161 were kept and wrapped in the beginning- and end-of-sequence tokens (1024 tokens at most). On
162 real cells, our implementation gives the same token sequences as the MaxToki tokenizer, except
163 that genes with exactly tied count-to-median values can appear in swapped order.

164 We recorded, for each gene in a panel, its hidden state at that gene’s token position and averaged
165 it over the cells of one cell type. The main MaxToki-217M extraction used layers 0, 4, 8 and 11.
166 A separate extraction of all 12 hidden states (the embedding output and all 11 transformer layers)
167 with a 4000-gene panel, on the same cells, gave the full depth profile. Four design choices each
168 fix a failure we measured in a pilot. (1) Pairwise contexts, not the all-context intersection: in the
169 main extraction only 137 genes are count-balanced in all 12 cell types, many of them broadly
170 expressed housekeeping genes (the single-cell analogue of the “stopwords” that are most context-
171 sensitive in language models [9]), whereas each pair of cell types shares a median of 1391 count-
172 balanced genes. The panel contains genes with at least 25 occurrences in at least two cell types
173 (the 6000 most frequent such genes). (2) Cell-level split halves: each cell was assigned to one
174 of two partitions, so every per-gene mean is estimated twice from disjoint cells and gives an
175 honest noise floor. (3) An occurrence cap: a per-gene mean has a standard error proportional
176 to $1/\sqrt{n}$, and unequal counts inject context-dependent noise into any interaction term, so at most
177 50 occurrences were accumulated per (gene, cell type, partition), over a shuffled cell order. A
178 (gene, cell type) entry is used only if it reached the cap in both partitions (“count-balanced”). (4)
179 Per-dimension z-scoring: before any cosine, every hidden dimension was standardised over all
180 count-balanced entries, to remove the inflation by a few dominant (“rogue”) dimensions that a
181 scalar anisotropy correction misses [13].

182 Other extractions used the same procedure with these settings. Scaling pair: MaxToki-217M

183 and MaxToki-1B on the same 600 cells per cell type with a 5000-gene panel (cap 50), at layers 2
 184 and 4 of MaxToki-217M and layers 4 and 7 of MaxToki-1B, so that the models can be compared
 185 at matched relative depth (layer 4 of 11 = 0.36 versus layer 7 of 20 = 0.35; layer 2 of 11 = 0.18
 186 versus layer 4 of 20 = 0.20). Random-weights control: the same cells and the same 5000-gene
 187 panel as the MaxToki-217M scaling run, at layers 4 and 8. Cap-matched control: the first 20
 188 occurrences of each (gene, cell type, partition) in the main extraction’s forward pass, at layer 4.
 189 Setty data: cap 25, a floor of 15 occurrences in at least two clusters, a 5000-gene panel, layers 2
 190 and 4. scGPT: representations from an existing per-gene activation atlas of 3000 Tabula Sapiens
 191 cells (immune, kidney and lung) [34], in which each cell’s 1200 highest-expressed genes were
 192 given to scGPT with their log-normalised values passed to the value channel without scGPT’s
 193 standard binning; we reformatted it into the same layout: cell types with at least 100 cells (8 cell
 194 types), two partitions, cap 20, floor 12, 5000 genes. STATE: a preprocessed copy of each tissue
 195 that keeps about 4800 highly variable genes, with ambient-corrected counts; 600 cells per tissue
 196 (immune, kidney and lung), taken as up to 150 cells from each of the largest cell types until 600
 197 were reached (in practice the 4 largest per tissue); STATE fills its input with unexpressed genes
 198 as well, and we kept only positions holding expressed genes (count > 0); cell types with at least
 199 100 cells (9 cell types), cap 20, floor 12 (1840 genes).

200 **Two-way decomposition and contextualisation strength (EXCESS)**

201 For gene g in cell type c , we write $v(g, c) = \mu + a(g) + b(c) + E(g, c)$: a gene main effect $a(g)$
 202 (the gene’s identity), a context main effect $b(c)$ (a shift shared by all genes in that cell type,
 203 expected because attention pools over the cell), and a gene-specific interaction $E(g, c)$, the only
 204 term where a gene-specific response to context can appear.

205 For a cell-type pair (c_0, c_1) , the relative shift of gene g is $\delta(g) = [v(g, c_1) - v(g, c_0)] -$

206 $\text{mean}_g[v(g', c_1) - v(g', c_0)]$: the gene's own shift minus the average shift of all genes. We
 207 computed δ separately in the two cell partitions (δ_0, δ_1) and defined $\text{EXCESS} = \text{mean cos}(\delta_0(g),$
 208 $\delta_1(g)) - \text{mean cos}(\delta_0(g), \delta_1(\text{perm } g))$: the same gene's shift agreeing across independent cells,
 209 minus a gene-shuffled control. $\text{EXCESS} = 0$ means no gene-specific response to context.
 210 The means are pooled over all (gene, pair) entries from the cell-type pairs with at least 200
 211 count-balanced genes (150 in the cross-model and Setty analyses), which equals a mean over
 212 pairs weighted by their number of genes. 95% confidence intervals (CIs) come from 2000
 213 bootstrap resamples of genes, in which each resampled gene brings all of its pair entries. As a
 214 positive control we report the replication of the context main effect across partitions (the cosine
 215 between the two partitions' mean shifts).

216 **Anisotropy.** On the raw (unstandardised) representations we computed the mean cosine of
 217 random different-gene pairs (the anisotropy baseline [9]), same-gene self-similarity across cell
 218 types, and the share of variance on the first principal component (PC), and repeated these after
 219 per-dimension z-scoring.

220 **Rank-position control.** MaxToki encodes token position, and a gene's rank changes between
 221 cell types. We computed each gene's mean rank in each cell type over the cells that express it,
 222 using the same cells as the extraction. We split genes at the median absolute rank change into
 223 rank-stable and rank-moving halves and recomputed EXCESS in each, correlated the size of
 224 the shift with the rank change, and recomputed EXCESS after regressing the shift on the gene's
 225 mean rank in each of the two cell types and on its absolute rank change.

226 **Functional axes and reproducible interaction power**

227 A functional axis u is the unit vector separating two GO-defined gene classes in the gene main
 228 effect, $u \propto \text{mean}(a(g) \mid \text{class A}) - \text{mean}(a(g) \mid \text{class B})$, where $a(g)$ is the gene's z-scored

229 representation averaged over partitions and count-balanced cell types. Genes annotated to
 230 both poles were removed from both. The axes were nuclear versus surface/secreted (nucleus
 231 GO:0005634, chromatin GO:0000785 and DNA binding GO:0003677, versus plasma mem-
 232 brane GO:0005886, extracellular region GO:0005576 and extracellular space GO:0005615);
 233 mitochondrion (GO:0005739) versus cytoskeleton (GO:0005856); transcription (regulation
 234 of DNA-templated transcription GO:0006355 and DNA-binding transcription factor activity
 235 GO:0003700) versus transport and signalling (ion transport GO:0006811 and signalling receptor
 236 activity GO:0038023); and cell cycle (GO:0007049) versus cell differentiation (GO:0030154).
 237 Axis validity is the five-fold cross-validated area under the receiver operating characteristic
 238 (ROC) curve (AUC) with which $\square a(g), u \square$ separates held-out pole genes.

239 Reproducible interaction power measures how much context moves genes along u in a way
 240 that replicates across independent cells. For genes count-balanced in at least 9 cell types, we
 241 projected each partition's representation onto u , $q_p(g, c) = \square v_p(g, c), u \square$; regressed out the gene's
 242 mean rank in that cell type (linear and quadratic terms; "rank-controlled"); took the two-way
 243 interaction $I_p(g, c)$ (the projection minus its gene mean and context mean, plus the grand mean);
 244 and averaged $I_0(g, c) \cdot I_1(g, c)$ over all (gene, cell type) entries. Noise averages to zero in this
 245 product, so only modulation that reproduces across the two partitions survives. Functional-z
 246 is the z-score of this power against 300 axes built from random gene partitions of the same
 247 pole sizes. We report it at layers 4, 8 and 11, without and with the rank regression. A cell
 248 type's loading on an axis is the mean projection onto u of the z-scored layer-4 representations
 249 (averaged over partitions) of the genes count-balanced in that cell type. These gene sets differ
 250 between cell types and the loadings are not rank-controlled, so we use them only to describe the
 251 ordering of cell types.

Null models for functional organisation

Functional gene sets may be co-expressed, and co-expressed genes co-vary across cells whatever the model “understands”. We therefore compared each functional axis’s power with null axes whose two poles have the same sizes as the functional poles. Co-expression was the Pearson correlation of $\log_{10}(\text{counts per 10,000})$, computed from the raw counts over 3000 cells from each of the three Tabula Sapiens datasets, and the coherence of a gene set was its mean pairwise correlation.

1. **Random axes:** random genes (300 axes).
2. **Strongly co-expressed modules:** each null pole was the set of genes most correlated with a random seed gene (300 axes). These modules are much more co-expressed than the functional poles (Results), so this is a strong null rather than a matched one.
3. **Size- and co-expression-matched sets:** each null pole was built from the k genes most correlated with a random seed gene plus randomly chosen genes, with k chosen by bisection so that the pole’s coherence was within 10% of the functional pole’s (300 axes). We also traced power against the fraction of seed-co-expressed genes at fixed pole size (0 to 100%, 25 draws per step).
4. **Size- and tightness-matched sets:** the same construction with representation tightness (the mean pairwise cosine of the pole genes’ main effects $a(g)$) in place of co-expression (300 axes).

For each null, the empirical one-sided p is $(k + 1)/(N + 1)$, where k is the number of the $N = 300$ null axes with power at least as large as the functional axis; when no null axis reached it, p takes its smallest possible value, $1/301$. Empirical p values were used because the null power distributions are right-skewed.

Causal steering test

To test whether the model uses the functional direction, we added it to the residual stream at half of a cell’s gene positions and read the model’s own next-gene output logits at the other half. The direction was the raw (unstandardised) difference between the mean layer-4 representations of the nuclear-pole and surface-pole genes (poles made disjoint, as for the axes). It was added at the layer-4 representation (the output of the fourth transformer block). For each of 30 immune cells (at most 512 tokens), gene positions were randomly split into a steered half and a read half, so any effect at the read positions must travel through attention. The readout was the mean change in logit of nuclear-pole gene tokens minus that of surface-pole gene tokens (specificity). Strengths were 0.25, 0.5 and 1.0 times the mean residual-stream norm at that layer in one reference cell; we report 0.5 as the headline. Pushing toward $+u$ and toward $-u$ and taking the difference (the signed swing) cancels any effect that does not depend on the sign of the push. The control was a random direction of the same norm, drawn fresh for each cell. Significance was a one-sided sign test of functional swing greater than random swing across cells. The test was repeated at the layer-8 representation and for the mitochondrion/cytoskeleton and transcription/transport axes at layer 4. To put the push sizes in context, we measured how far context naturally moves genes along the same direction: the spread of each gene’s projection onto u across its count-balanced cell types.

Tests for context-specific gene function

Directional congruence. Along each functional axis, we decomposed each gene’s projection into a gene loading $f(g)$ (mean over contexts), a context loading $h(c)$ (mean over genes) and an interaction $e(g,c)$. If genes move toward their own functional pole specifically in cell types that share that character, e should follow $f \cdot h$. We estimated f and h in one partition and e in the other

(and vice versa), measured $\beta = \text{corr}(e, f \cdot h)$ on rank-residualised projections, and compared β with 300 null axes whose poles were matched to the functional poles on size and co-expression (as in null 3 above).

Placement near curated targets. For TFs count-balanced in at least 6 cell types with at least 15 curated targets in the gene panel, we asked (A) whether, in each cell type, the TF’s representation (with the context mean removed) is closer by cosine to the centroid of its curated targets than to the centroid of non-targets matched on co-expression with the TF (40 redraws), and (B) whether this excess closeness rises in cell types where the targets are more highly expressed (Spearman correlation across cell types with target activity, the negative of the targets’ mean rank). A cell type was used for a TF only if at least 8 of its targets were count-balanced there, and a TF needed at least 3 such cell types. Both questions were tested across TFs with sign tests.

Lineage transcription factors. We also attempted a test on 14 TFs with lineage-specific roles in immune cells (SPI1, IRF4, IRF8, GATA3, MAFB, BCL6, PRDM1, TBX21, RUNX1, RUNX3, BATE, MAF, BACH2 and ZEB2) [40], but few of them were sampled well enough in the panel (Results).

Cross-model, scaling and random-weights comparison

EXCESS and nuclear/surface functional-z were computed identically for every model at layer 4, and for the scaling pair also at matched relative depth, on each model’s own gene vocabulary (context pairs with at least 150 count-balanced genes). In this comparison, and in the Setty replication, functional-z was computed without the rank regression, on genes count-balanced in at least $\lceil n/2 \rceil$ of the n cell types (4 of 9 for STATE and the Setty clusters), against 200 random-partition axes. Because the occurrence cap differs between architectures (50 for MaxToki, 20 for scGPT and STATE), we also report MaxToki-217M with cap 20.

Link to next-gene prediction

For 350 Tabula Sapiens immune cells with at least 60 expressed vocabulary genes (at most 512 tokens), and up to 20 target positions per cell from the sixth gene onward, we compared the model's log-probability of the true gene given the cell's own preceding genes with that given the same number of preceding genes from another cell (a chimeric context). The donor cell was drawn at random among cells with at least as many genes whose preceding genes did not include the target gene, so that the chimeric context never repeats the target. The per-gene context benefit (log-probability difference, averaged over occurrences; genes with at least 3 occurrences) was correlated with the gene's per-gene context movement, the mean over its count-balanced cell types (at least 4) of $\|z(g,c) - \text{mean}_c z(g,c)\|$ in the z-scored layer-4 space (a per-gene quantity, distinct from EXCESS). We used a partial Spearman correlation controlling for log gene frequency, with a 95% CI from 2000 bootstrap resamples of genes.

Statistics and reproducibility

All analyses are deterministic (random seed 0) and run on cached extractions. Empirical p values are one-sided unless stated. Multiple tests were corrected within each family: the four steering configurations and the four curated-target tests (static closeness and its modulation, at layers 4 and 8) with a Bonferroni correction, and the six directional-congruence tests (three axes, layers 4 and 8; only positive β counted) with a Holm correction. Each analysis has its own script: input encoding `ctx_tokenise.py`; extraction `ctx_extract_maxtoki.py` (and `_scgpt`, `_state`, `_devel`, `_random`); `ctx_polysemy.py`, `ctx_layer_curve.py` and `ctx_anisotropy.py` (contextualisation by layer and anisotropy); `ctx_position_confound.py` (rank control); `ctx_independent.py` (Setty replication); `ctx_functional_axes.py`, `ctx_celltype_loadings.py`, `ctx_coexpr_null_v2.py`,

344 `ctx_coexpr_null_v3.py` and `ctx_tightness_null.py` (functional axes and nulls);
345 `ctx_causal.py` (steering); `ctx_directional_probe.py`, `ctx_curated_targets.py` and
346 `ctx_switcher_test.py` (tests for context-specific function); `ctx_cross_model.py` (cross-
347 model comparison); `ctx_prediction_link.py` (prediction link); and `ctx_figures.py`
348 (figures). Each writes a result file `results/ctx_*.json`, from which every number, table and
349 figure in this paper is taken. The runs used Python 3.12 with PyTorch 2.12.1, transformers
350 5.13.0 and NumPy 2.4 on Apple-silicon hardware.

351 **Use of large language models**

352 An artificial intelligence (AI) assistant (Claude, Anthropic) was used to help write and run the
353 analysis code, check references, and draft and edit the manuscript text. The author designed
354 the study, reviewed the code, results and text, and takes full responsibility for the content. The
355 assistant is not an author.

356 **Results**

357 All analyses below use MaxToki-217M unless a model is named. We first establish Level 1
358 (contextualisation is real, not rank position, replicable, partly learned and weakly linked to pre-
359 diction, and one functional direction is read by the model when pushed), then report the Level
360 2 tests.

361 **Genes are contextualised gene-specifically, most strongly in the first layers**

362 The gene-specific context response is large and highly reproducible (Table 1, Fig. 1). At layer 4
363 of the main extraction, EXCESS is +0.758: the same gene's relative shift agrees at cosine +0.759
364 across independent cells, while a different gene's agrees at +0.001, so the effect is gene-specific.

365 The context main effect replicates at +0.996, which shows that the measurement works. Layer
366 0 (the context-free embedding output) returns exactly 0.000, the check that the pipeline reports
367 nothing when there is nothing. The full depth scan shows where the response arises: it is highest
368 at layer 1 (+0.847) and layer 2 (+0.835) and then falls to +0.651 at layer 11 (Fig. 1). Context
369 is therefore integrated in the first one or two transformer layers. Deeper layers keep most of it
370 (90% of the peak at layer 4 and 84% at layer 8), while the gene-specific shift becomes somewhat
371 less reproducible toward the output.

Table 1. Gene-specific context response at the four layers of the main MaxToki-217M extraction

Layer	EXCESS (same			Main-effect
	– different)	Same gene	Different gene	replication
0 (embedding)	+0.000	+0.000	+0.000	—
4	+0.758	+0.759	+0.001	+0.996
8	+0.711	+0.711	+0.000	+0.996
11	+0.649	+0.650	+0.001	+0.997

372 EXCESS is the agreement of the same gene’s relative context shift across two independent halves
373 of the cells, minus the agreement for a different gene, pooled over 66 cell-type pairs (a median
374 of 1391 count-balanced genes per pair). Gene-level bootstrap 95% CIs are within ± 0.003 . The
375 full depth profile is shown in Fig. 1.

376 **Metric validity (anisotropy).** These representations are anisotropic, as expected [9]: the mean
377 cosine of random different-gene pairs is +0.20 to +0.62 at layers 4–11 (raw). Same-gene self-
378 similarity across cell types falls with depth (+0.92 $\rightarrow$ +0.86 $\rightarrow$ +0.69 at layers 4, 8 and 11),
379 so genes move more between cell types in upper layers, the single-cell echo of Ethayarajh’s
380 finding. Together with the EXCESS profile, this means that upper layers move genes more but

less reproducibly. Per-dimension z-scoring brings anisotropy to about 0.00 while keeping genes distinguishable (corrected self-similarity +0.88/+0.65/+0.62), and the first principal component carries only 0.07 of the variance at layer 11, so no single dimension dominates the cosines.

The effect is not the gene's rank position

MaxToki reads a cell as a rank-ordered gene list and encodes token position, so a gene ranking 5th in one cell type and 500th in another is an obvious confound. It is not the driver (Fig. 2). The gene-specific response is about the same for rank-stable genes (+0.749) and rank-moving genes (+0.767) at layer 4, and 99% of it survives regression on rank (+0.749); 98% and 97% survive at layers 8 and 11. The size of a gene's shift correlates only weakly with its rank change ($\rho = 0.26$).

The effect replicates on an independent dataset

On the Setty CD34+ bone-marrow data, the lineage in which a recoverable manifold was found in scGPT [62], with nine differentiation clusters rather than adult cell types as contexts, the phenomenon replicates: EXCESS is +0.713 at layer 2 and +0.713 at layer 4 (gene-shuffled null +0.0036 and +0.0039), and functional-z is +5.0 and +5.0 for the nuclear/surface axis and +16.1 and +8.8 for the mitochondrion/cytoskeleton axis. Using spliced counts only gives the same picture (EXCESS +0.702 and nuclear/surface functional-z +9.3 at layer 4). Neither the contextualisation nor its functional organisation is specific to Tabula Sapiens or to adult cell types.

Context moves genes along functional axes, as co-expression does

Context moves genes along interpretable functional axes far more than along random directions (Table 2). At layer 4, functional-z (rank-controlled in parentheses) is +16.6 (+13.3) for nuclear versus surface/secreted genes, +8.7 (+6.4) for mitochondrion versus cytoskeleton and +2.9 (+4.3) for transcription versus transport. The cell-cycle versus differentiation axis, which separates its poles only weakly (validity AUC 0.57), gives +2.0 (+0.9). The organisation persists in deeper layers for the two strongest axes. Cell types order sensibly along the nuclear/surface axis: lymphocytes sit at the nuclear end (CD8 T +4.7, CD4 T +4.4, B +3.9) and alveolar epithelial cells and macrophages at the surface end (alveolar type 1 -3.6, type 2 -3.2, macrophage -2.8).

Table 2. Functional organisation of context modulation, by axis and layer (MaxToki-217M)

Axis	Poles (genes)	Validity AUC			
		(layer 4)	Layer 4	Layer 8	Layer 11
Nuclear / surface	1800/1188	0.77	+16.6 (+13.3)	+7.8 (+6.5)	+6.1 (+6.0)
Mitochondrion / cytoskeleton	675/165	0.78	+8.7 (+6.4)	+12.7 (+8.4)	+10.4 (+6.9)
Transcription / transport	326/89	0.69	+2.9 (+4.3)	+1.0 (+1.3)	+0.8 (+0.8)
Cell cycle / differentiation	145/167	0.57	+2.0 (+0.9)	+2.9 (+1.7)	+2.6 (+1.9)

410 Functional-z is the z-score of the axis's reproducible interaction power against 300 axes built
 411 from random gene sets of the same pole sizes, without (and, in parentheses, with) regression on
 412 the gene's rank. Validity AUC is the cross-validated area under the ROC curve with which the
 413 axis separates held-out pole genes.

414 The key control asks whether this organisation exceeds co-expression (Fig. 3). Functional gene
 415 sets can be co-regulated, and co-regulated genes co-vary across cells whatever the model "un-
 416 derstands". The functional poles are more co-expressed than random genes, but far less than
 417 co-expression modules (mean pairwise correlation 0.028 and 0.042 for the nuclear and surface
 418 poles, against 0.025 for random gene sets). Against null gene sets matched on size and this
 419 average co-expression, no functional axis exceeds the null: the nuclear/surface power (0.50)
 420 sits inside the matched null (mean 0.40, 95th percentile 0.68; $p = 0.316$), and $p = 0.375$ and
 421 0.083 for the other two axes. Modules built from the genes most co-expressed with a seed gene,
 422 3–11 times more co-expressed than the functional poles, reach higher power on average (nu-
 423 clear/surface null mean 0.67; $p = 0.701$, 0.757 and 0.804). At fixed pole size, 66 of 175 null
 424 axes with 0–100% seed-co-expressed genes reached the nuclear/surface power. At layer 4 the
 425 organisation is not a by-product of compact gene sets in representation space: it exceeds sets
 426 matched on size and representation tightness ($p = 0.007$, 0.023 and 0.013). At layer 8 it does
 427 not ($p = 0.233$, 0.060 and 0.100). It far exceeds random axes (none of 300 reached it; $p = 1/301$
 428 for nuclear/surface). So the functional organisation of the contextualisation is no stronger than
 429 that of gene sets with the same co-expression.

The nuclear/surface direction is read by the model, but not every functional direction is

In the steering test (Fig. 4), adding the nuclear/surface direction to half of a cell's genes moves the model's own logits at the untouched half with the sign of the push: nuclear-gene logits rise for a push toward the nuclear pole (specificity +0.085) and fall for a push toward the surface pole (−0.097), a signed swing of +0.182 against +0.001 for a random direction of the same norm. The swing is +0.079, +0.182, +0.184 at strengths 0.25, 0.5 and 1.0 and exceeds the random control in 22 of 30 cells ($p = 0.0081$; Bonferroni-adjusted 0.032). The effect is stronger at the layer-8 representation (swing +0.232, 30 of 30 cells, $p = 9.3 \times 10^{-10}$). The direction therefore propagates through attention and changes the model's predictions.

This does not generalise to every functional axis. The mitochondrion/cytoskeleton direction has no significant effect at strengths 0.25 and 0.5 (swing +0.024, +0.067, +0.269; at 0.5, 15 of 30 cells, $p = 0.57$); at strength 1.0 it exceeds the random control in 20 of 30 cells ($p = 0.049$, Bonferroni-adjusted 0.20), the same as the nuclear/surface direction at that strength. The transcription/transport direction has a reversed effect that grows with strength (swing −0.139, −0.268, −0.678): the functional swing exceeds the random one in only 2 of 30 cells (two-sided sign test $p = 8.7 \times 10^{-7}$). We have no explanation for the reversal; it shows that an axis that separates genes is not automatically read in the expected direction. The pushes are also large compared with natural variation: the middle strength (7.6 units) is about 3 times the median range of a gene's projection on the nuclear/surface direction across cell types (2.62). The test therefore shows that the model can read this axis, not that the natural, context-driven changes along it are used.

No context-specific gene function beyond co-expression

Two direct tests for context-appropriate gene function find no context-specific signal beyond co-expression. (1) Directional congruence: genes do not move toward their own functional pole specifically in cell types that share that character more than matched null axes do. At layer 4, β is +0.16, +0.10 and +0.23 for the three axes, all positive but within the upper tail of size- and co-expression-matched null axes, whose means are near zero ($p = 0.116, 0.086$ and 0.053 ; none significant after Holm correction over 6 tests; at layer 8, β is $-0.16, -0.11$ and $+0.09$). The null spread of β (standard deviation 0.08–0.13) means the test could detect only clear congruence. (2) Placement near curated targets: across 26 TFs, at layer 4 a TF's representation is slightly closer to the centroid of its curated targets than to that of co-expression-matched non-targets (mean cosine excess +0.021; 20 of 26 TFs positive; sign test $p = 0.005$, Bonferroni-adjusted over 4 tests 0.019). This closeness does not rise significantly in the cell types where the targets are more active (mean $\rho = +0.073$; 17 of 26 TFs positive; $p = 0.08$, Bonferroni-adjusted 0.34), and at layer 8 neither effect is significant (static $p = 0.279$; modulation $\rho = +0.135$, $p = 0.08$). So the model places TFs modestly near their curated targets beyond what co-expression with the TF predicts, but not in a context-appropriate way. A third attempt, on 14 lineage TFs, is uninformative rather than negative: only 1 of them (RUNX1) was sampled well enough, because these regulators are weakly expressed in the cells we used.

Contextualisation as a model metric: architecture, scale, and learned versus architectural

Because it does not depend on resolving gene function, contextualisation can serve as a model metric (Table 3, Fig. 5). Three findings stand out. (1) Every model contextualises genes strongly (EXCESS 0.71–0.83), but contextualisation does not grow consistently with model size. At

475 matched relative depth, MaxToki-1B gives +0.767 (layer 7) against +0.740 for MaxToki-217M
 476 (layer 4), and +0.767 (layer 4) against +0.831 (layer 2). Functional organisation is not higher
 477 in the larger model either (functional-z +8.1 against +14.3, and +18.4 against +18.4). (2) Func-
 478 tional organisation differs sharply between architectures: the rank-based MaxToki models give
 479 $z = 8.1-18.4$ (Table 3), scGPT +3.2 and STATE +1.1. STATE moves genes strongly in context
 480 (EXCESS +0.828) but in no functionally coherent direction; its gene tokens are fixed protein
 481 embeddings rather than learned from co-expression. The models differ in more than tokenisa-
 482 tion (training data, objective, gene panel and context sets), so this comparison does not isolate
 483 the cause. (3) The phenomenon is partly architectural and partly learned. An untrained Max-
 484 Toki of the same architecture, run on the same cells and gene panel, keeps about 60% of the
 485 trained contextualisation (EXCESS +0.441 against +0.740) but only about a third of its func-
 486 tional organisation (functional-z +4.7 against +14.3). Fixed random embeddings and attention
 487 mixing give a reproducible, gene-specific shift on their own; training strengthens it and aligns it
 488 with functional axes. The untrained MaxToki’s functional-z (+4.7) is higher than that of trained
 489 scGPT (+3.2) and STATE (+1.1), so part of what this metric measures comes from MaxToki’s
 490 rank-based input and architecture rather than from training.

Table 3. Contextualisation strength and functional organisation across models

Model	Layer				EXCESS	Functional- z
	(relative depth)	Contexts	Genes	Width		
scGPT	4 (0.42)	8	5000	512	+0.705	+3.2
STATE	4 (0.31)	9	1840	2048	+0.828	+1.1
(SE-600M)						

Model	Layer		Contexts	Genes	Width	EXCESS	Functional- z
	(relative depth)						
MaxToki- 217M	2 (0.18)		12	5000	1232	+0.831	+18.4
MaxToki- 217M	4 (0.36)		12	5000	1232	+0.740	+14.3
MaxToki- 1B	4 (0.20)		12	5000	2304	+0.767	+18.4
MaxToki- 1B	7 (0.35)		12	5000	2304	+0.767	+8.1
MaxToki- 217M, random weights	4 (0.36)		12	5000	1232	+0.441	+4.7

EXCESS measures contextualisation strength; functional-z measures organisation along the nu-
clear/surface axis against 200 random-gene axes (no rank regression). Relative depth is the
number of transformer blocks applied divided by the model's total number of blocks: for Max-
Toki, layer k follows block k ; for scGPT and STATE, layer 4 follows the fifth block (5/12 and
5/16). The MaxToki rows use the same 600 cells per cell type and the same 5000-gene panel
(cap 50); scGPT and STATE come from their own extractions (cap 20), so their EXCESS should
not be ranked against MaxToki's (Limitations).

Contextualisation is weakly aligned with what the model uses for prediction

If contextualisation serves the autoregressive objective, the genes the model contextualises most should be those whose next-gene prediction gains most from real context. That is what we find. Real context helps prediction a great deal: across 2266 comparisons for 558 genes, the mean benefit is +12.0 nats (median +12.9; a nat is a natural-log unit of probability), so the true gene is roughly 160,000-fold more probable under its real context than under another cell's. The per-gene size of this benefit tracks how much the gene moves between cell types: the partial Spearman correlation, controlling for gene frequency, is +0.161 (95% CI +0.083 to +0.239). Genes whose representations depend more on context tend, weakly, to be genes whose prediction depends more on context.

Discussion

The results characterise how SCFMs represent genes in context and what that representation carries. On the positive side, MaxToki reshapes gene representations gene-specifically and reproducibly, most strongly in its first one or two layers. Training strengthens the reshaping (an untrained model keeps about 60% of its strength and a third of its functional organisation); the reshaping is weakly aligned, gene by gene, with how much the model's predictions depend on context; and in the MaxToki models it follows functional axes. To our knowledge this is the first measurement for single-cell models of the "how contextual are contextualised representations" question [9], and the comparison of models gives an interpretability metric that separates the three architectures we tested, though not the two MaxToki sizes consistently.

What the reshaping carries is, as far as our tests can tell, co-expression. The functional organisation of the context changes is no stronger than that of gene sets matched to the functional sets on

size and average co-expression, and modules of strongly co-expressed genes produce more; we did not test directly whether a representation built from expression alone reproduces it. The two direct tests for context-specific gene function are negative. The one signal beyond co-expression is static: TFs sit slightly closer to their curated targets than to co-expression-matched non-targets at layer 4, but not more so in the cell types where those targets are active. The steering result fits the same picture. The model reads the nuclear/surface direction, but the pushes tested (3.8–15.2 units) are 1.4 to 6 times a gene’s typical natural range along it, so the test does not show that natural context changes are used; and two other functional directions are not consistently read, or are read in reverse. The architecture comparison does not contradict this: STATE, whose gene tokens are fixed protein embeddings, contextualises genes strongly but without functional organisation, and scGPT, also trained on expression, shows little; the comparison cannot separate tokenisation from the other differences between the models.

One possible reason for this boundary is that an expression-only model sees only which genes are expressed and their rank order, and its objective rewards predicting co-expressed genes. The alignment between contextualisation and the benefit of real context for prediction is consistent with that reading, although the test measures only how far a gene moves, not in which direction, so it does not show that the model moves genes toward their co-expressed partners. Some context-dependent function acts at the protein level, as in moonlighting proteins [39], and leaves no expression signature. Other forms, such as cell-type-specific enhancer choice by lineage-determining TFs [40], act through target-gene expression and could in principle be visible; our tests did not detect them. Whether multi-omic, sequence-aware or larger expression models reach Level 2 is not tested here.

Limitations

Most analyses were run on MaxToki-217M only: the rank control, the Setty replication, the steering test, the co-expression and tightness nulls, the prediction link and both Level 2 tests. MaxToki-1B, scGPT and STATE were tested only for contextualisation strength and nuclear/surface functional organisation. Cross-architecture EXCESS is confounded by the occurrence cap (20 for scGPT and STATE versus 50 for MaxToki) and by width. A MaxToki-217M run with the same cap of 20 gives EXCESS +0.620 and functional-z +28.0, so the functional-z gap between architectures is not caused by the cap, but EXCESS values should not be ranked across architectures. The functional-z comparison uses non-identical context sets (12, 9 and 8 cell types) and, for STATE, a smaller gene panel (1840 genes) and only 14 of 36 cell-type pairs with enough count-balanced genes; we did not re-run it on a common context set. STATE was given only about 4800 highly variable genes per tissue (ambient-corrected counts), which is likely one reason its panel holds only 1840 genes and few cell-type pairs could be scored. The rank control was run only for MaxToki; scGPT and STATE also receive each gene's expression level, so part of their EXCESS may reflect the gene's own expression change rather than its context. The scGPT representations come from an existing extraction in which log-normalised values were given to scGPT's value channel without its standard binning, so its absolute activations are off its training distribution; our analyses use contrasts that largely cancel this, but the scGPT numbers should be read as relative to MaxToki, not as calibrated values. The scaling comparison has two model sizes only.

Most analyses use layer 4, not the peak layers 1–2; the co-expression nulls were run only at layer 4 and the tightness null at layers 4 and 8. The steering direction is the context-averaged nuclear-minus-surface direction, a gene-class direction rather than the context-specific change itself, and it was compared only with a random push, not with co-expression-matched direc-

tions. The co-expression nulls match average pairwise co-expression; power also depends on how co-expression is distributed within a gene set, and we did not run the more direct test of whether a representation built only from expression (for example, a factorisation of the same cells) reproduces the functional organisation. The Setty data cover a pre-filtered set of genes, unlike the full raw counts of Tabula Sapiens. The Level 2 negatives are evidence of a ceiling, not proof of absence: some context-dependent function is invisible to an expression-only model, and the directional test could detect only clear congruence. Functional axes are built from direct annotations in one annotation file; the conclusions hold across three axes, but specific z-scores depend on how the poles are defined.

Conclusions

Single-cell foundation models build contextual gene representations that are a genuine object of study: a gene's vector is reshaped by its cellular context in a gene-specific and reproducible way that training strengthens, most strongly in the first layers, weakly aligned with how much the model's predictions rely on context, and different between architectures. To our knowledge, this is the first systematic characterisation of the phenomenon for single-cell models. The functional organisation of the contextual change gives a simple metric that separates the three architectures we tested. In MaxToki, controls matched on co-expression also bound what the contextualisation carries: its functional organisation is no stronger than that of gene sets with the same co-expression, and two direct tests for context-specific gene function are negative. Reading a gene's changed role in a new cell type, the use that would make these representations a hypothesis-generation tool, is not reachable with the expression-only model tested here and will likely require models trained on more than expression.

List of abbreviations

AI: artificial intelligence; AUC: area under the receiver operating characteristic curve; CI: confidence interval; CLP: common lymphoid progenitor; DC: dendritic cell; ESM2: Evolutionary Scale Modeling 2; GO: Gene Ontology; HSC: haematopoietic stem cell; PC: principal component; ROC: receiver operating characteristic; SCFM: single-cell foundation model; TF: transcription factor.

Declarations

Ethics approval and consent to participate

Ethics approval and consent to participate were not required for this study, which is a secondary analysis of publicly available, de-identified human single-cell RNA-sequencing data (Tabula Sapiens [45] and Setty et al. [50]) and involved no contact with participants. The original studies obtained ethics approval and informed consent, as described in those publications. No new human data or samples were collected.

Consent for publication

Not applicable.

Availability of data and materials

The datasets analysed during the current study are available in the following public repositories: the Tabula Sapiens immune, kidney and lung datasets in CZ CELLxGENE Discover [47–49]; the human CD34+ bone-marrow data of Setty et al. [50], as distributed with scVelo, in figshare [52]; the MaxToki checkpoints in Hugging Face [41]; the scGPT whole-human checkpoint [42];

the STATE SE-600M model in Hugging Face [43]; the Geneformer gene dictionaries in Hugging Face [61]; the GEARS gene-to-GO annotation file in Harvard Dataverse [56]; TRRUST v2 [58]; and DoRothEA [60]. All analysis code, the result files from which every number, table and figure is computed, the figure code and the list of the 20,000 immune cells used are available in the gene-context repository [63]. The MaxToki, random-weights, Setty and STATE representation tensors (about 9 GB) are not deposited because of their size; they can be regenerated from the public inputs with the provided extraction scripts and are available from the corresponding author on reasonable request. The scGPT representations were reformatted from an activation atlas produced in earlier work [34]; that atlas is available from the corresponding author on reasonable request. No new experimental data were generated.

Software availability:

- Project name: gene-context
- Project home page: <https://github.com/Biodyn-AI/gene-context>
- Archived version: git tag bmc-submission-v1 of the project repository
- Operating system(s): platform independent (developed and run on macOS, Apple silicon)
- Programming language: Python 3.12
- Other requirements: NumPy, SciPy, scikit-learn, h5py and matplotlib; for the extraction scripts, PyTorch (2.2 or later) and transformers (4.43 or later), and for STATE the Arc Institute state package and anndata; versions used: PyTorch 2.12.1, transformers 5.13.0, NumPy 2.4
- License: MIT
- Any restrictions to use by non-academics: none

Competing interests

The author declares that he has no competing interests.

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Authors' contributions

IK conceived and designed the study, wrote the software, carried out the analyses, interpreted the results and wrote the manuscript. The author read and approved the final manuscript.

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825 **Figures**

826 **Figure 1. Gene-specific context response from the embedding through all 11 layers of**
 827 **MaxToki-217M.** Agreement of the same gene’s relative context shift across independent halves
 828 of the cells (blue; full depth scan with a 4000-gene panel) against the gene-shuffled null (grey).

Open diamonds mark the four layers of the main 6000-gene extraction (Table 1). Layer 0, the context-free embedding output, is exactly zero; the response is highest at layer 1 (+0.847) and falls to +0.651 at layer 11. Values are pooled over 66 cell-type pairs per layer; gene-level bootstrap 95% CIs are narrower than the markers.

Figure 2. The gene-specific context response is not explained by rank position. EXCESS at layers 4, 8 and 11 for genes whose rank changes little between cell types (rank-stable, below the median rank change), genes whose rank changes a lot (rank-moving, above the median), and all genes after regressing the shift on the gene's ranks and rank change. The response is about the same for rank-stable and rank-moving genes, and 99%, 98% and 97% of it survives the rank regression at layers 4, 8 and 11.

Figure 3. Functional organisation compared with random and co-expression nulls. For each of the three functional axes with well-separated poles (layer 4; the cell-cycle/differentiation axis is omitted), the red marker and line show the functional axis's rank-controlled reproducible interaction power. Boxes show 300 null axes each, all with the functional poles' sizes: random genes (grey); gene sets matched on size and average co-expression (light blue); and modules of the genes most co-expressed with a random seed gene (dark blue), which are 3–11 times more co-expressed than the functional poles. Boxes show the interquartile range and median; whiskers reach the most extreme null value within 1.5 times the interquartile range (values beyond are not drawn). Annotated p values are empirical one-sided values against the matched null and against the strongly co-expressed modules. The functional axes are far above random axes but within both co-expression nulls.

Figure 4. The nuclear/surface direction is read by the model. Signed swing (the change in nuclear-minus-surface logit specificity for a push toward the nuclear pole minus that for a push toward the surface pole) at the untouched half of each cell's genes, when the nuclear/surface di-

rection is added at the layer-4 representation of the other half, as a function of steering strength (multiples of the mean residual-stream norm). Blue: functional direction (mean $\pm$ standard error over 30 cells). Grey: random direction of the same norm. The mitochondrion/cytoskeleton direction has no significant effect at strengths 0.25 and 0.5, and the transcription/transport direction a reversed one (Results).

Figure 5. Contextualisation as a model metric across architecture, scale and training. Left: contextualisation strength (EXCESS). Right: functional organisation (nuclear/surface functional-z against random-gene axes, without rank regression); the dashed line marks $z = 3$, a reference level for a clearly non-random effect. Rows: scGPT and STATE (SE-600M) at layer 4, each from its own extraction (occurrence cap 20); MaxToki-217M at layer 4 and MaxToki-1B at layer 7 (matched relative depth, 0.36 and 0.35), on the same 600 cells per cell type (cap 50); and MaxToki-217M with random weights (grey). Every trained model contextualises strongly, but only the MaxToki models show strong functional organisation; random weights give weaker contextualisation and weak functional organisation. Because the caps differ, EXCESS should not be ranked across architectures (Limitations).